

ACHINTYA RAI

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EDUCATION

University of California, San Diego
Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028
GPA: 3.9/4.0

WORK EXPERIENCE

Adobe

Software Engineering Intern

San Jose, CA

June 2026 – Sept. 2026

- Profiled **60.6M**+ payment events in Databricks data pipelines, flagging **86,387** duplicate records and a **47%** null timestamp rate to establish the source of truth for downstream monitoring.
- Designed a time-series anomaly detector, selecting a machine learning model over **30**+ countries.
- Developed MLOps pipelines scheduling weekly retraining and hourly inference in production.

VALT Health

Founding Software Engineer

Remote

Aug. 2025 – Feb. 2026

- Engineered a Flask backend handling **100k**+ datapoints, unifying data across **20**+ device sources.
- Implemented async batch processing in GCP for a **35%** speed gain and **60%** fewer redundant writes.
- Automated environment setup and data seeding via Bash scripts, cutting testing time up to **80%**.

Decklar (formerly Roambee)

Software Engineering Intern

Sunnyvale, CA

June 2024 – Aug. 2024

- Developed real-time shipment monitoring modules for **Amazon, Coca-Cola, and Samsung**, optimizing Python data processing pipelines for a **15%** capacity boost across **1M**+ weekly events.
- Implemented advanced data structures for analytics modules, yielding **20%** faster reporting.
- Enhanced workflow efficiency by **15%** via Panels-based UI refinements and user testing.

RESEARCH

Graph Theory & Neural Architecture

Undergraduate Research Fellow (NSF REU)

San Diego, CA

March 2026 – Present

- Engineered a Python pipeline to analyze LLM embedding manifolds (Llama-2, DeepSeek-V2), building weight-matrix extraction utilities that cut **500GB** checkpoints to **13GB** localized manifolds.
- Optimized preprocessing for a **30x** runtime reduction, enabling analysis on **8GB RAM** systems.
- Applied Rent's Rule and Louvain clustering to raise ICN density from **0.18%** to **13.25%**.

Applications of Multi-Agent Systems

Undergraduate Research Fellow (UC Scholars)

San Diego, CA

June 2025 – Aug. 2025

Multi-agent AI Chess Tutor | *Gemini API, MCP, Django, React, SQLite*

- Built an AI chess analysis tool with Django, React (TypeScript), and SQLite via Agile sprints.
- Architected multi-agent workflows that cut hallucinations **80%** while running **83%** faster.
- Engineered custom **MCP** servers and **7 REST API** endpoints validated by **15**+ chess coaches.

PROJECTS

SquaredUpChess | *React, TypeScript, Django, WebSockets, AWS, Docker* 

Aug. 2025 – Dec. 2025

- Engineered a full-stack platform on **AWS** with **20+** **Django REST API** endpoints and WebSockets.
- Established CI/CD with automated testing to deploy to production at **99.9%** reliability.

NanoGPT | *Python, PyTorch, Transformers, Neural Networks* 

Mar. 2026

- Trained a character-level transformer with **100k**+ parameters in Python/PyTorch from scratch.

Watchdog | *Python, Gemini API, browser-use, Asyncio, SQLite* 

April 2026

- Engineered an agentic browser safeguard using the browser-use SDK to autonomously monitor tabs.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, TypeScript, JavaScript, Node.js, SQL (Postgres/SQLite), Bash

Development: React, Django, FastAPI, Flask, WebSockets, Asyncio, REST APIs, CI/CD, PyTest, JUnit

AI & Cloud: PyTorch, TensorFlow, LLMs, MCP, GitHub Copilot, Claude Code, AWS, GCP, Docker, Linux